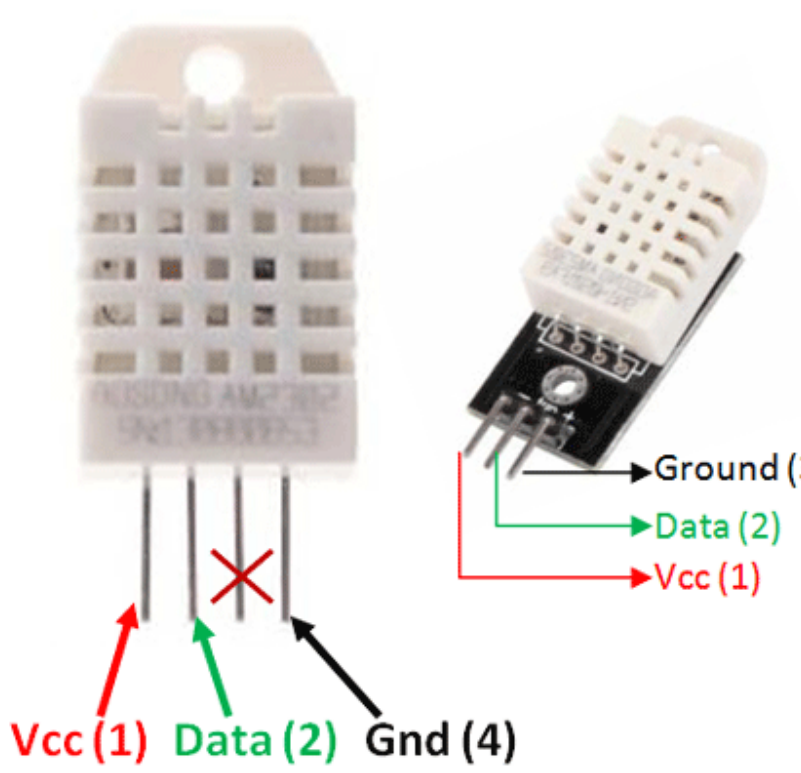
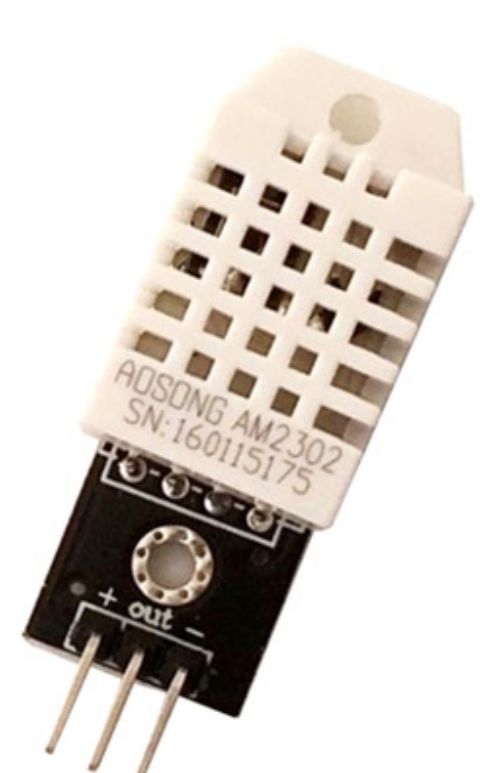


DHT22 – Temperature and Humidity Sensor

19 April 2018 – 0 Comments



DHT22 Temperature Sensor Module		DHT22 Sensor Pinout	
No:	Pin Name	Description	
For DHT22 Sensor			
1	Vcc	Power supply 3.5V to 5.5V	
2	Data	Outputs both Temperature and Humidity through serial Data	
3	NC	No Connection and hence not used	
4	Ground	Connected to the ground of the circuit	
For DHT22 Module			
1	Vcc	Power supply 3.5V to 5.5V	
2	Data	Outputs both Temperature and Humidity through serial Data	
3	Ground	Connected to the ground of the circuit	

DHT22 Specifications

- Operating Voltage: 3.5V to 5.5V
- Operating current: 0.3mA (measuring) 60uA (standby)
- Output: Serial data
- Temperature Range: -40°C to 80°C
- Humidity Range: 0% to 100%
- Resolution: Temperature and Humidity both are 16-bit
- Accuracy: ±0.5°C and ±1%

Note: The **DHT22 datasheet** can be found at the bottom of the page

DHT22 Equivalent Temperature Sensors

DHT11, AM2302, SHT71

Other Temperature Sensors

Thermocouple, TMP100, LM75, DS18B20, SHT15, LM35DZ, TPA81, D6T

Difference between DHT22 Sensor and Module

The DHT22 sensor is the successor of the **DHT11 module**, it can either be purchased as a sensor or as a module. Either way the performance of the sensor is same. The sensor will come as a 4-pin package out of which only three pin will be used whereas the module will come with just three pins as shown in the **DHT22 pinout** above.

The only difference between the sensor and module is that the module will have a filtering capacitor and pull-up resistor inbuilt, and for the sensor you have to use them externally if required.

The module is slightly costly than the DHT11, but it has a higher measuring range and slightly better accuracy .

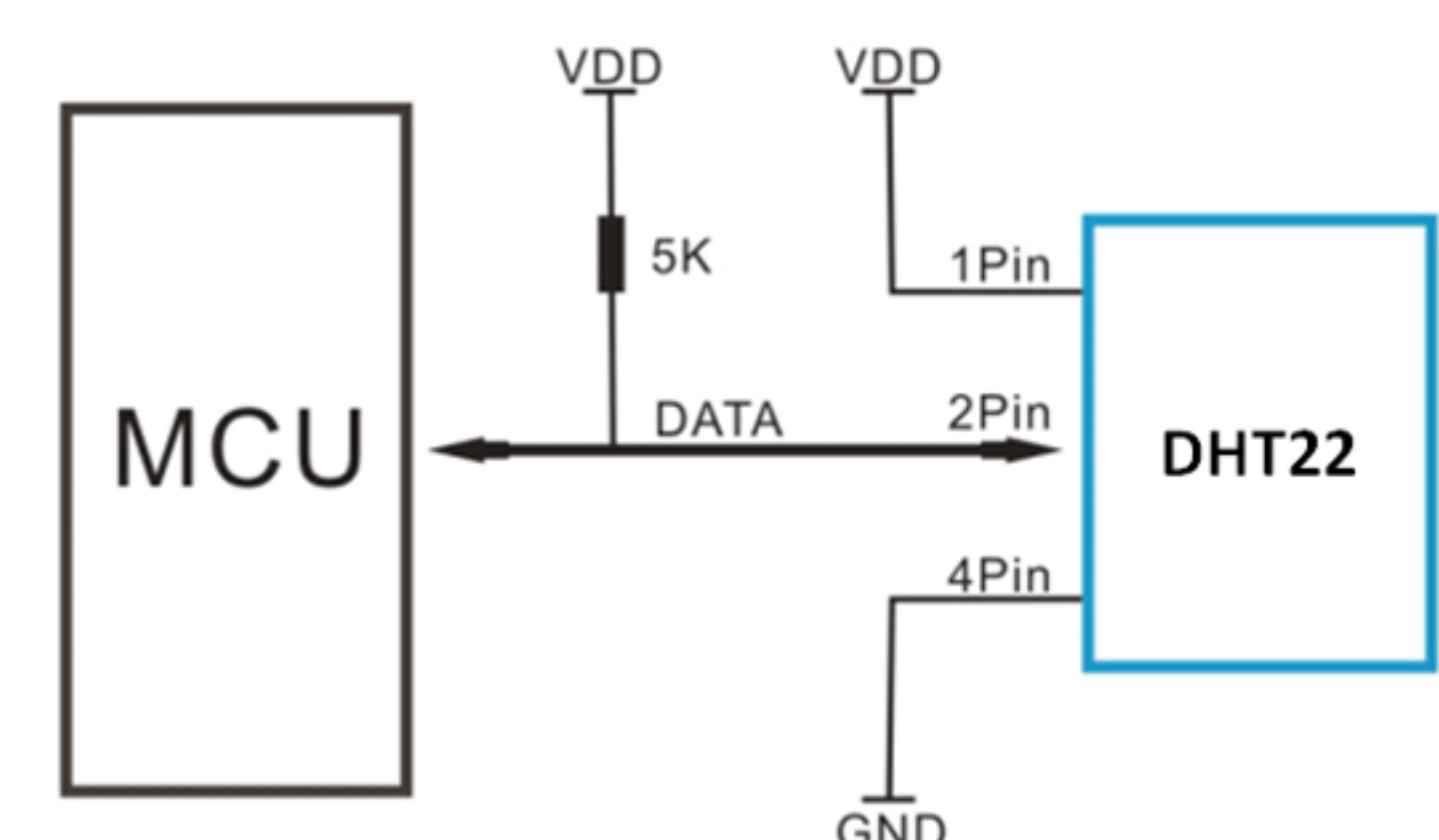
Where DHT22 Sensor is Used

The **DHT22** is a commonly used Temperature and humidity sensor. The sensor comes with a dedicated **NTC** to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor is also factory calibrated and hence easy to interface with other microcontrollers.

The sensor can measure temperature from **-40°C to 80°C** and humidity from **0% to 100%** with an accuracy of ±1°C and ±1%. So if you are looking to measure in this range then this sensor might be the right choice for you.

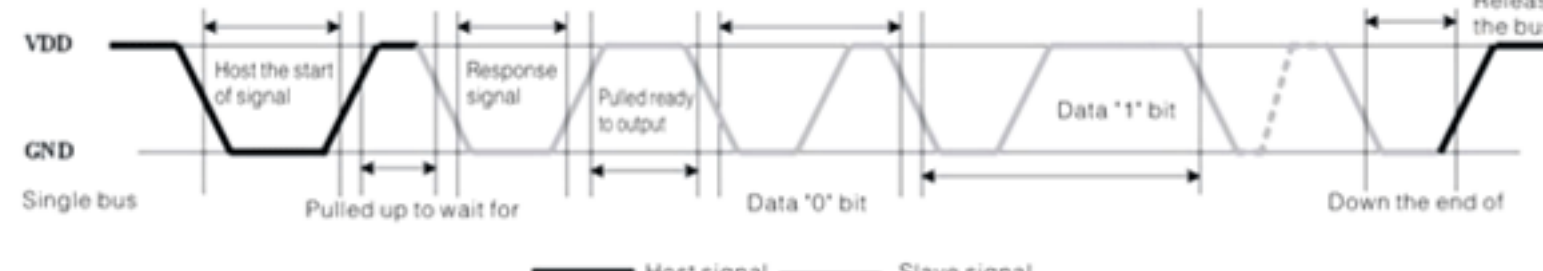
How to use DHT22 Sensor

The **DHT22 Sensor** is factory calibrated and outputs serial data and hence it is highly easy to set it up. The connection diagram for this sensor is shown below.



As you can see the data pin is connected to an I/O pin of the MCU and a 5K pull up resistor is used. This data pin outputs the value of both temperature and humidity as serial data. If you are trying to **interface DHT22 with Arduino Uno** then there are ready made libraries for it which will give you a quick start.

If you are trying to interface it with some other MCU then the datasheet given below will come in handy. The output given out by the data pin will be in the order of 8bit humidity integer data + 8bit the Humidity decimal data + 8 bit temperature integer data + 8bit fractional temperature data + 8 bit parity bit. To request the DHT11 module to send these data the I/O pin has to be momentarily made low and then held high as shown in the timing diagram below.

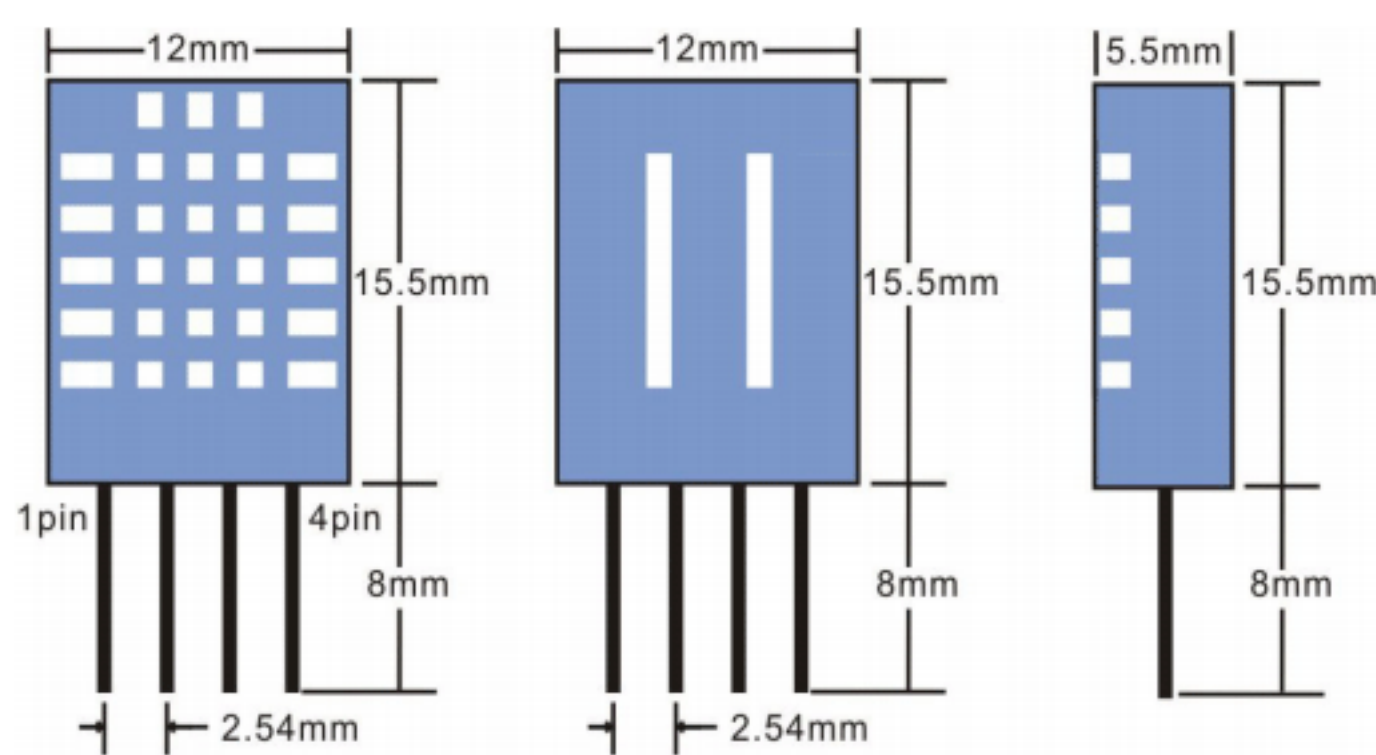


The duration of each host signal is explained in the datasheet, with neat steps and illustrative timing diagrams

Applications

- Measure temperature and humidity
- Local Weather station
- Automatic climate control
- Environment monitoring

2D – model of the sensor



Component Datasheet

SEN0137 DHT22 Sensor Datasheet

Where to Buy

Click here to view more results for SEN0137

	Distributor ↕	Part Number ↕	Stock ↕	
	element14	SEN0137	140	Buy Now
	Farnell	SEN0137	140	Buy Now
	Mouser Electronics	SEN0137	121	Buy Now
	DigiKey	SEN0137	97	Buy Now
	Verical	SEN0137	57	Buy Now
	Arrow Electronics	SEN0137	57	Buy Now
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	RS Americas	SEN0137	0	Buy Now
	Newark	SEN0137	0	Buy Now

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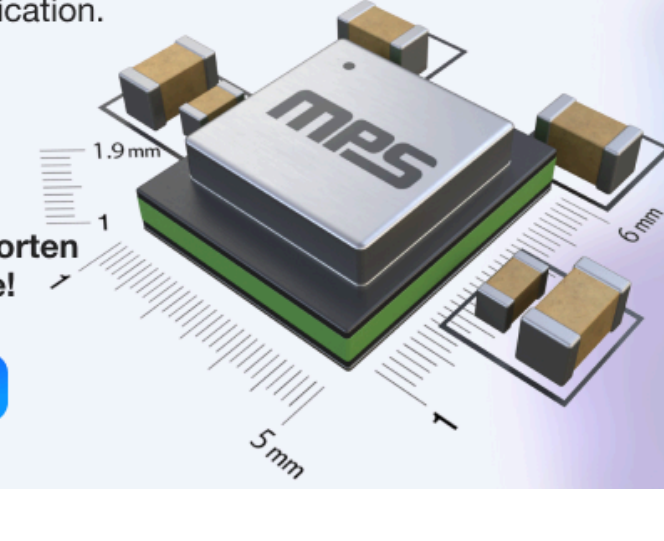


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